

DATA ANALYTICS BOOTCAMP

IronHack May 2026

DATA ANALYTICS

Analyzing What Drives Quality in an Iron Processing Plant

```
print("Author: Adriana Alves")
```

4,097 hours from one industrial plant

DATASET SCOPE

4,097

hours of plant operation

20-sec
Hourly

process readings (pH, airflow, dosing)

lab values (iron & silica content)

What was measured

Process inputs:

pH · starch flow · amina flow · ore pulp flow and density · 7 air-flow columns · 7 column levels

Lab outputs:

Iron content in & out · silica content in & out

Time scope:

Continuous operating hours, with no roster, maintenance or volume records.

What can the plant's data tell us?

Five questions guided the analysis, each grounded in operational reality.

01

TIME PATTERNS

How does quality change across hours, days and time-of-day windows?

02

QUALITY DISTRIBUTION

What share of operating hours produce top vs standard vs low quality?

03

STABILITY

Are top periods more stable: running more smoothly on the controls operators adjust?

04

PERSISTENCE

How long do top quality periods last? How fast does quality recover after a drop?

05

COMMERCIAL IMPACT

How much is the revenue during top quality hours?

From raw readings to tested hypotheses



TECH STACK

MySQL · Python · pandas · scipy · Jupyter · matplotlib · seaborn · Tableau

Output quality across all 4,097 hours

49.9

%

TOP

2,044 hours

32.4

%

STANDARD

1,329 hours

17.7

%

LOW

724 hours

TAKEAWAY Half of all hours qualify as top quality, but nearly one in five lands in penalty territory.

How often quality is at its best, by time of day?

Top Quality % by time of day

MORNING

06:00–11:59



AFTERNOON

12:00–17:59



EVENING

18:00–23:59



NIGHT

00:00–05:59



WHAT I SEE

Nighttime hours (evening + night) sit at the high end.
Daytime hours sit at the low end.

Day of week: I found no meaningful pattern between days of the week and weekends.

Months: Only 6 months of data, not enough to find a real trend.

Three hypothesis were tested

Each hypothesis maps to a project goal and a specific statistical test.

H1

STABILITY

Are top quality hours running more steadily on the controls operators adjust?

TEST · Mann-Whitney U

H2

TIME-OF-DAY

Is top quality associated with time-of-day buckets?

TEST · Chi-squared

H3

PERSISTENCE

Do top quality periods last longer than lower quality streaks?

TEST · Mann-Whitney U

Are top quality hours steadier?

THE CLAIM

Top quality output is associated with steadier operating conditions during that hour.

THE TEST

Mann-Whitney U comparing within-hour variability (standard deviation) of process variable.

VARIABLE

p-VALUE

SIGNIFICANT?

pH variability

0.022

✓ yes, at $\alpha = 0.05$

Air-flow variability (avg)

0.042

✓ yes, at $\alpha = 0.05$

Level variability (avg)

0.077

✗ no

Amina flow variability

0.544

✗ no

Starch flow variability

0.593

✗ no

VERDICT

Partial support: 2 of 5 variables differ significantly. Stability is not a strong single signal of top quality output.

Is better quality tied to the time of day?

CLAIM *Output quality varies systematically with the time of day and the pattern reflects more than just random noise.*

p-value

0.030

Effect size (Cramer's V)

0.041

VERDICT *Real but weak. With 4,097 hours, tiny effects show as significant. Time of day matter but only a little.*

Does top quality results last?

TYPICALLY RUNS FOR

3

hours

max 61h

388 distinct premium streaks identified

Mann-Whitney U • one-sided

$p < 0.001$

Effect size (rank-biserial)

0.246

Small-to-medium, which is a real, moderate effect.

VERDICT

Supported. Top quality output is a somewhat persistent state and not a one-hour fluke.

How I turned quality into revenue

REVENUE PER TONNE (\$)

$$P_{\text{base}} + (\text{Fe} - 65) \cdot \text{VIU}_{\text{Fe}} - \max(0, \text{Si} - 2.0) \cdot \text{Penalty}_{\text{Si}}$$

=

\$120.9

Parameters in place

P-BASE

\$122

per tonne

Platts 65% Fe reference price

IRON PREMIUM

\$2.20

per 1% Fe

Value of each extra % iron

CONTAMINANT PENALTY

\$2.00

per 1% Si

Deduction per % above 2.0%

What this analysis cannot say

ONE PLANT

4,097 hours from a single plant. Findings may not generalise.

NO CAUSATION

All results are associations, can't pinpoint a causal relationship.

NO VOLUMES

No tonnage in dataset. Revenue uses an assumed production rate.

NO ROSTERS

No operator, shift or maintenance records, those effects are invisible here.

SIZE EFFECT

With 4,097 hours, tiny effects show as significant. Effect sizes matter more than p-values.

LAB CADENCE

Quality lab values are hourly. Within-hour quality drift is not captured.

■ KEY TAKEAWAYS

01

Top quality is a frequent state, not a one-off.

Good runs hold typically for **3 hours** and **50%** of the output is top quality.

02

Most quality drops recover quickly.

After dropping out of top quality, the plant usually recovers within about 2 hours. But a minority of stretches last far longer, up to a few days in the worst cases. *Those rare long stretches are where most lost time accumulates and are worth catching early.*

03

Nearly 1 in 5 hours produces a final output that will suffer price penalty.

Applying the way iron ore buyers price for quality, about 18% of operating hours fall into the penalty band. Once production volume is known, this becomes a dollar figure.

04

What did not was related to the quality output?

Two common assumptions didn't hold: the plant running more steadily did *not* reliably mean better quality and quality is only marginally better at night. *Knowing what doesn't drive quality narrows where to look next.*